Mruthulan Senthil Nathan

Singapore | mruthulansenthilnathan@gmail.com | mruthulan.com
linkedin.com/in/mruthulan | github.com/mru34
Full-stack developer internship | Available late Mar. 2027

EDUCATION

Singapore Polytechnic Expected May 2028
Diploma in Information Technology

SELECTED PROJECTS

Better Call Bhai Apr. 2026 | Private pilot
Independent barbershop booking platform | *Node.js, Express, SQLite* | Project

- Identified a local barber handling appointments through WhatsApp, built a booking prototype independently and secured the owner's participation in a private pilot.
- Built and deployed the service and time-slot booking flow, Express API and owner admin page on Render, with a WhatsApp link to booking details.
- Prevented duplicate booking slots with a SQLite uniqueness constraint and rejected past-date bookings through server-side validation.

SignalBridge - Champion, Dell InnovateDash 2026 Jun. 2026
Youth support platform | *Next.js, FastAPI, PostgreSQL* | *Team of four* | Project | Code

- Built the youth chat, sign-in and dashboard in Next.js, connecting authentication and saved conversations to the team's FastAPI backend.
- Implemented a handoff preview and consent flow so young people could review the prepared note and submit permission for worker review.
- Extended support to Discord with thread-based intake, new-user onboarding and database migrations; added pytest integration tests for message routing and intake.

MEANT Aug. - Sept. 2026
Assistive communication app | *2nd Runner-Up, Dell InnovateFest 2026* | Project

- Designed colour-coded symbol boards, vocabulary search and onboarding, allowing users to compose replies and find words beyond the AI suggestions.
- Presented the team's application, which received the S$3,000 polytechnic-category prize and was featured in Tamil Murasu.

Boss Breaker Dec. 2025 - Feb. 2026
Solo coursework | *Wellness game* | *JavaScript, Express, MySQL* | Code

- Built the frontend, REST API and MySQL database for challenges, boss raids and inventory; implemented server-side game calculations, bcrypt hashing and JWT-protected routes.

TECHNICAL SKILLS

- **Web:** TypeScript, JavaScript, React, Next.js, Node.js, Express, HTML/CSS
- **Backend and data:** Python, FastAPI, REST APIs, SQL, PostgreSQL, MySQL, SQLite
- **Tools:** Git, GitHub, pytest, Render

LEADERSHIP

Secretary, Youth Harmony Chapter subcommittee Jun. 2026 - Present
SP ACER, Singapore Polytechnic May 2025 - Present

AWARDS

- **Champion, Autodesk Singapore Hackathon 2026** - Built and delivered the presentation for KnowCad, the team's customer service AI assistant.